

All orders asymptotics and asymptotic inversion for A292507

Research note

1 October 2026

Abstract

We prove an all-orders Poincaré expansion for the binomial transform of the number of partitions without parts equal to one. This gives a precise additive version of the logarithmic conjecture recorded for OEIS A292507. A finite symbolic coefficient algorithm, six computed correction terms, an all-orders reversion algorithm, and rigorous residual and integer-threshold bounds are supplied. The leading binomial-smoothing principle is already recorded in OEIS A218481 and is not claimed as new. The report establishes the mathematics of the expansion; its limited literature search does not establish publication priority.

Provenance, status and reading conventions

[write] **Added 2 October 2026, batch 77.** *Provenance.* This report was built from one manuscript: batch 77, manuscript 01 of the repository’s incoming-reports channel, archive `A292507_report_and_code.zip` (main file `a292507_asymptotics.tex`, 474 lines, with a ten-page PDF that is not shipped), arrival commit `096ee7b87`, placed in `4f11bc9c0`. From the abstract to the bibliography the text is the delivered manuscript, unchanged except for the label prefix `bpt:` and the notes marked [write]; section, theorem and equation numbers are the manuscript’s own. *Pin.* The delivery names no `ProveIt` commit and continues no repository path; at placement no repository file mentioned A292507, A218481, A002865 or A292622. *Status.* An unrefereed research note; the delivery names no author or tool (its author line reads “Research note”). Nothing in it is formalized in Lean or Rocq, and its place in the collection confers no formal status. On 2 October 2026, on scratch copies, the delivered `validate_inverse.py` passed, and it and the recorded run `reproduce.py -order 6 -max-n 10000` wrote `inverse_verification.json` and `verification.json` identical to the shipped files apart from line endings; an independent exact computation of a_n from (1) confirmed the ten initial terms below and the values of c_1, c_2 numerically: at $n = 6000$ the scaled residual $n(a_n/(C2^n n^{-3/2} e^{\beta\sqrt{n}}) - 1 - c_1 n^{-1/2})$ is 15.13, against $c_2 + c_3 n^{-1/2} = 15.10$. These replays check the transcription and the implementation, not the proofs.

[write] **Reading conventions.** The manuscript reuses several letters. No symbol of the delivered text was renamed; the table fixes the readings and the tempting false ones.

symbol	meanings, by section
c, c_j	$c = \pi\sqrt{2/3}$ is the Hardy–Ramanujan exponent of $p(k)$; the c_j are the correction coefficients of (3), with $c_0 = 1$. The repository’s partition chapter (see the note after Lemma 2.1) calls c “ λ ”.
λ	here $\lambda = \beta/2 = \pi/(2\sqrt{3})$, the exponential tilt of the binomial variable Z (Sections 3–4); <i>not</i> $\pi\sqrt{2/3}$.
$A, \mathcal{A}_M$	$A = 1/(4\sqrt{3})$; $\mathcal{A}_M(x)$ is the real model (28).
C	the leading constant $C = (\pi/6)e^{\pi^2/24}$ of (2) everywhere <i>except</i> in (18), where “ $Cn^{3/2}$ ” is an unspecified constant; $C_{s,r}$ in (17) is another. $\mathcal{C}_n$ is the central event.
P	$P(x)$ is the partition approximation (8); $P_j(z)$ are the Taylor polynomials of Lemma 3.1, and $P_j(\partial_s)$ the operators of (25); $\mathbb{P}$ is probability.
b	$b(k) = p(k) - p(k-1)$; $b_r = \lfloor s^{2r} \rfloor \log \cosh s$ in (24).
D, d_j	$D(t)$ of (11), $\tilde{D}(t) = D(\sqrt{2}t) = \sum d_j t^j$; D_M in the proof of Theorem 6.1 is an existence constant.
$B_\bullet$	B_0, B_r, B'_r and B_M are unrelated existence constants; in particular the B_M of (22) is <i>not</i> the bracket radius B_M of (40).
F	$F_q(t)$ (10) and $\tilde{F}_q(t)$ (26) are partition-side functions; $F_M(x) = \log \mathcal{A}_M(x)$ (29) is the inverse-side model.
$E, \mathbb{E}$	$E(h, z)$ is the exponent (21); $\mathbb{E}$ is expectation.
K	$K \sim \text{Bin}(n, 1/2)$ in Sections 3–4; K, K', K'' in the proof of Theorem 5.2 are constants.
M, M_n	M is a truncation order; $M_n(s)$ is the moment-generating function (14).
L	L_n is the normaliser of Section 3; in Section 5 $L = \log X$ and $L_0 = \log C$.
t, v	t is the argument of $D(t)$, evaluated at $k^{-1/2}$; in Section 4 it is the formal variable of a truncated exponential with coefficients v_j ; in the proof of Theorem 5.2 $t = X^{-1/2}$, and v, v_j are the normalised unknown and its truncations.
T, r	T is a Chernoff threshold, $T_J(X)$ the inverse approximant (35); r is a moment exponent, a cumulant index or a derivative order, and $r = (M+1)/2$ in Theorem 6.1.

1 Definition and main result

Let $p(k)$ be the ordinary partition function, with $p(0) = 1$ and $p(-1) = 0$, and put

$$b(k) = p(k) - p(k-1), \quad a_n = \sum_{k=0}^n \binom{n}{k} b(k). \quad (1)$$

Here $b(k)$ counts partitions without ones: removing one part of size one bijects the remaining partitions with partitions of $k-1$. Consequently

$$a_n = [x^n](1+x)^n \prod_{m \geq 2} (1-x^m)^{-1},$$

by expanding the polynomial and then replacing k by $n-k$. This agrees with the formula recorded in A292507 [1]. The initial terms are 1, 1, 2, 5, 13, 33, 82, 201, 488, 1176.

Throughout the report, define

$$c = \pi\sqrt{2/3}, \quad A = \frac{1}{4\sqrt{3}}, \quad \beta = \frac{c}{\sqrt{2}} = \frac{\pi}{\sqrt{3}}, \quad \lambda = \frac{\beta}{2}, \quad C = \frac{\pi}{6}e^{\pi^2/24}. \quad (2)$$

All asymptotics are for positive integers $n \rightarrow \infty$, with the truncation order fixed.

Theorem 1.1. *There is a uniquely determined sequence of real constants $(c_j)_{j \geq 0}$, with $c_0 = 1$, such that for every integer $M \geq 0$,*

$$a_n = C 2^n n^{-3/2} e^{\beta \sqrt{n}} \left(\sum_{j=0}^M c_j n^{-j/2} + \mathcal{O}_M(n^{-(M+1)/2}) \right). \quad (3)$$

The finite algorithm in Section 4 computes every c_j . In particular,

$$c_1 = -\frac{17\beta}{12} - \frac{3}{\beta} - \frac{\beta^3}{32}, \quad (4)$$

$$c_2 = \frac{9\beta^8 + 1008\beta^6 + 32416\beta^4 + 156672\beta^2 + 55296}{18432\beta^2}, \quad (5)$$

$$c_3 = -\frac{27\beta^{10} + 5400\beta^8 + 389088\beta^6 + 11153152\beta^4}{5308416\beta} - \frac{85893120\beta^2 + 112803840}{5308416\beta}. \quad (6)$$

Taking logarithms is legitimate because the normalized expression tends to one. Thus, with the formal coefficients $\ell_j = [t^j] \log(\sum_{r \geq 0} c_r t^r)$,

$$\log a_n = n \log 2 + \beta \sqrt{n} - \frac{3}{2} \log n + \log C + \sum_{j=1}^M \ell_j n^{-j/2} + \mathcal{O}_M(n^{-(M+1)/2}). \quad (7)$$

In particular $\ell_1 = c_1$ and $\ell_2 = c_2 - c_1^2/2$. This proves more than the recorded relation $\log a_n \sim n \log 2 + \pi \sqrt{n/3} - \frac{3}{2} \log n$: (7) specifies the subleading terms additively. A ratio asymptotic between two expressions of size n would not by itself determine their $\sqrt{n}$ and $\log n$ terms.

2 The partition input and finite differences

The only non-elementary input needed for the proof is the following established partition estimate. For real $x > 1/24$, write

$$P(x) = \frac{A e^{c\sqrt{x-1/24}}}{x - 1/24} \left(1 - \frac{1}{c\sqrt{x-1/24}} \right). \quad (8)$$

Nemes' Lemma 2.2 and its displayed identification of the convergent coefficient series give, for integers $k \geq 1$ [4],

$$|p(k) - P(k)| \leq \frac{A}{k} e^{c\sqrt{k}/2}. \quad (9)$$

This is the positive-exponential part of the first partition term, with a genuine exponentially smaller remainder. Merely differentiating a leading equivalent for $p(k)$ would not justify its finite difference.

To extract that difference without losing accuracy, define functions analytic at $t = 0$ by removable continuation:

$$F_q(t) = \frac{\exp(c(\sqrt{1-qt^2} - 1)/t)}{1 - qt^2} \left(1 - \frac{t}{c\sqrt{1-qt^2}} \right), \quad (10)$$

$$D(t) = \frac{2}{ct} (F_{1/24}(t) - F_{25/24}(t)). \quad (11)$$

All square roots take their branch equal to one at zero. The numerator of (11) vanishes to first order, so D is analytic and $D(0) = 1$.

Lemma 2.1. *For any fixed $0 < \eta < c/2$, as integer $k \rightarrow \infty$,*

$$b(k) = \frac{Ac}{2} k^{-3/2} e^{c\sqrt{k}} \left(D(k^{-1/2}) + \mathcal{O}(e^{-\eta\sqrt{k}}) \right). \quad (12)$$

Moreover $0 \leq b(k) \leq B_0 e^{c\sqrt{k}}$ for all $k \geq 0$ and some constant B_0 .

Proof. By direct substitution, $P(k) = Ak^{-1} e^{c\sqrt{k}} F_{1/24}(k^{-1/2})$ and $P(k-1) = Ak^{-1} e^{c\sqrt{k}} F_{25/24}(k^{-1/2})$. Subtracting gives the main term in (12). The sum of the two errors in (9) is $\mathcal{O}(k^{-1} e^{c\sqrt{k}/2})$ for $k \geq 2$. After division by $k^{-3/2} e^{c\sqrt{k}}$ it is $\mathcal{O}(\sqrt{k} e^{-c\sqrt{k}/2}) = \mathcal{O}(e^{-\eta\sqrt{k}})$. The final bound follows from $0 \leq b(k) \leq p(k)$, (8)–(9), and enlargement of the constant for finitely many small k . $\square$

[write] **Added 2 October 2026, batch 77: the partition input in the repository.** The function P of (8) is the dominant Rademacher sector $P_{1,+}$ of the partition chapter of the repository’s transseries volume (Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex, chapter p3:sec:top, display p3:eq:dominant-in-N and p3:thm:exact-sectors), whose $N = n - 1/24$, $\lambda = \pi\sqrt{2/3}$ and $\mu = \lambda\sqrt{N}$ are this article’s $x - 1/24$, c and $c\sqrt{x - 1/24}$. That chapter takes Rademacher’s convergent series as quoted input (p3:thm:rademacher). Its exact relative factorisation p3:prop:relative and its finite- K tail bound p3:thm:tail with $K = 1$ give, for $\mu \geq 2$ (true for every $k \geq 1$), $|p(k)/P(k) - 1| \leq 3e^{-2\mu} + \frac{8}{3}\mu^2 e^{-\mu/2}$, hence $|p(k) - P(k)| = \mathcal{O}(e^{c\sqrt{k}/2})$. This is weaker than (9) by a factor of order k , which the proof of Lemma 2.1 absorbs because $\eta < c/2$ is strict; so the volume supplies an in-repository alternative to Nemes’ estimate. This derivation is the writer’s, made when the report entered the collection, and is not part of the delivered manuscript. The same chapter describes the secondary sectors $P_{k,\pm}$ that the last paragraph of Section 8 leaves to separate analysis; nothing there treats the binomial transform.

For example, Taylor expansion gives

$$D(t) = 1 + \left(-\frac{13c}{48} - \frac{3}{c} \right) t + \mathcal{O}(t^2).$$

Define $\tilde{D}(t) = D(\sqrt{2}t) = \sum_{j \geq 0} d_j t^j$. Then

$$d_0 = 1, \quad d_1 = -\frac{13\beta}{24} - \frac{3}{\beta}. \quad (13)$$

3 Uniform smoothing and the tails

Let $K \sim \text{Bin}(n, 1/2)$, $h = n^{-1/2}$, and $Z = (2K - n)/\sqrt{n}$. The binomial transform is exactly $a_n = 2^n \mathbb{E}b(K)$. Its moment-generating function is

$$M_n(s) = \mathbb{E}e^{sZ} = \cosh(hs)^{1/h^2}. \quad (14)$$

For real s , $M_n(s) \leq e^{s^2/2}$: integrate $\tanh u \leq u$ for $u \geq 0$, use evenness, and exponentiate. Chernoff’s inequality then gives

$$\mathbb{P}(|Z| > T) \leq 2e^{-T^2/2}. \quad (15)$$

For every fixed real s and nonnegative integer r ,

$$\sup_n \mathbb{E}[e^{sZ} (1 + |Z|)^r] < \infty. \quad (16)$$

Indeed, a polynomial is bounded by a constant times $e^{|Z|}$, and the two signs of Z are bounded using $M_n(s+1) + M_n(s-1)$. Cauchy–Schwarz and (15) also give

$$\mathbb{E}[e^{sZ} (1 + |Z|)^r \mathbf{1}_{\{|Z| > T\}}] \leq C_{s,r} e^{-T^2/4}. \quad (17)$$

Use the fixed-fraction central event $\mathcal{C}_n = \{|Z| \leq (2h)^{-1}\}$, which is equivalent to $n/4 \leq K \leq 3n/4$. Concavity of the square root gives, for every $0 \leq K \leq n$,

$$c\sqrt{K} \leq c\sqrt{n/2} + \lambda Z.$$

Set

$$L_n = \frac{Ac}{2}(n/2)^{-3/2}e^{c\sqrt{n/2}} = \frac{\pi}{6}n^{-3/2}e^{\beta\sqrt{n}}.$$

Lemma 2.1 and (17) imply

$$\frac{\mathbb{E}[b(K)\mathbf{1}_{\mathcal{C}_n^c}]}{L_n} \leq Cn^{3/2}\mathbb{E}[e^{\lambda Z}\mathbf{1}_{\mathcal{C}_n^c}] = \mathcal{O}(n^{3/2}e^{-n/16}). \quad (18)$$

This handles $K = 0, 1$ as well and does not apply a large- k partition approximation at the endpoints.

On $\mathcal{C}_n$, formula (12) gives

$$\frac{b(K)}{L_n} = e^{\lambda Z} \left(H(h, Z) + \mathcal{O}(e^{-\kappa/h}) \right) \quad (19)$$

for some $\kappa > 0$, uniformly there, where

$$H(h, z) = e^{E(h, z)}(1 + hz)^{-3/2} \tilde{D}\left(\frac{h}{\sqrt{1 + hz}}\right), \quad (20)$$

$$E(h, z) = \frac{\beta}{h}(\sqrt{1 + hz} - 1) - \frac{\beta z}{2} = -\frac{\beta h z^2}{2(1 + \sqrt{1 + hz})^2}. \quad (21)$$

The error in (19) follows since $K \geq n/4$ and $(1 + hz)^{-3/2}$ is bounded. Both exponentials from (12) are handled by (21); its sign is especially useful.

Lemma 3.1. *Let $P_j(z) = [h^j]H(h, z)$, with $H(0, z) = 1$. These are polynomials in z . For every fixed M , there are $h_0 > 0$ and B_M such that*

$$\left| H(h, z) - \sum_{j=0}^M h^j P_j(z) \right| \leq B_M h^{M+1} (1 + |z|)^{2M+2} \quad (22)$$

when $0 < h \leq h_0$ and $|hz| \leq 1/2$.

Proof. Apply the one-variable Taylor theorem in the nonnegative variable h , with z held fixed. For every intermediate $0 \leq u \leq h$, $1 + uz \geq 1/2$ and $E(u, z) \leq 0$. Differentiating the last expression in (21) shows, for $r \geq 1$,

$$|\partial_u^r E(u, z)| \leq B_r (1 + |z|)^{r+1}.$$

The product formula for derivatives of e^E therefore bounds its r th derivative by $B'_r(1 + |z|)^{2r}$: a product with derivative orders summing to r has at most r factors, and $e^E \leq 1$. Derivatives of $(1 + uz)^{-3/2}$ of order r are bounded by a constant times $(1 + |z|)^r$. Put $w(u) = u(1 + uz)^{-1/2}$. Its derivatives of positive order r are bounded by a constant times $(1 + |z|)^{r-1}$, using $|uz| \leq 1/2$. Also $|w(u)| \leq \sqrt{2}h_0$. Choose h_0 inside a compact analytic neighborhood of $\tilde{D}$; all its required derivatives there are bounded. The chain and product rules now bound the $(M + 1)$ st derivative of H by the right-hand polynomial in (22), up to a constant. Taylor's integral remainder proves the assertion. Its coefficients are polynomials because every fixed-order formal expansion of (20) is polynomial in z . $\square$

Combining (19), Lemma 3.1, and (16), then removing the central restriction from each polynomial using (17), gives

$$\frac{\mathbb{E}b(K)}{L_n} = \sum_{j=0}^M h^j \mathbb{E}[e^{\lambda Z} P_j(Z)] + \mathcal{O}_M(h^{M+1}). \quad (23)$$

Thus the growing central range causes no loss of powers in the error: its Taylor remainders are integrated against uniformly bounded, exponentially weighted moments.

4 A finite coefficient algorithm

Let $b_r = [s^{2r}] \log \cosh s$ and define polynomials Q_v by the formal identity

$$\sum_{v \geq 0} h^v Q_v(s) = \exp \left(\sum_{r \geq 2} b_r h^{2r-2} s^{2r} \right). \quad (24)$$

For example $Q_0 = 1$, $Q_1 = 0$, $Q_2 = -s^4/12$, $Q_3 = 0$, and $Q_4 = s^6/45 + s^8/288$. Define the finite differential operator $P_j(\partial_s)$ by replacing each z^r in $P_j(z)$ with ∂_s^r . Then all coefficients in Theorem 1.1 are

$$c_m = e^{-\lambda^2/2} \sum_{j+v=m} P_j(\partial_s) (e^{s^2/2} Q_v(s)) \Big|_{s=\lambda}. \quad (25)$$

Only coefficients through degree m in h are needed in (20) and (24); the procedure is finite for every m .

Completion of the proof of Theorem 1.1. The analytic function $h^{-2} \log \cosh(hs)$ has a removable singularity at $h = 0$. On any fixed complex neighborhood of $s = \lambda$, its Taylor expansion converges uniformly for small h . Hence (14) has expansion

$$M_n(s) = e^{s^2/2} \left(\sum_{v=0}^M h^v Q_v(s) + \mathcal{O}_M(h^{M+1}) \right).$$

The remainder and any fixed number of s -derivatives have the same order, by Cauchy's formula on a slightly larger compact neighborhood. Since $\mathbb{E}[e^{sZ} P_j(Z)] = P_j(\partial_s) M_n(s)$ exactly, insert this expansion in (23) and collect powers of h . Its leading coefficient is $e^{\lambda^2/2}$. Multiplication by $2^n L_n$ gives (3) with $C = (\pi/6)e^{\lambda^2/2}$ and (25). Uniqueness follows by subtracting two finite expansions and considering their first differing coefficient. $\square$

For direct implementation it is convenient to avoid square roots in the symbolic coefficient field. Define

$$\tilde{F}_q(t) = \frac{\exp(\beta(\sqrt{1-2qt^2} - 1)/t)}{1-2qt^2} \left(1 - \frac{t}{\beta\sqrt{1-2qt^2}} \right). \quad (26)$$

Then $\tilde{D}(t) = (\tilde{F}_{1/24}(t) - \tilde{F}_{25/24}(t))/(\beta t)$. Compute this through order M (the numerator through order $M+1$), and use

$$\sum_{j \geq 0} h^j P_j(z) = \exp \left(\beta \sum_{r \geq 2} \binom{1/2}{r} h^{r-1} z^r \right) \sum_{j \geq 0} d_j h^j (1 + hz)^{-(3+j)/2}. \quad (27)$$

Every operation lies in $\mathbb{Q}[\beta, \beta^{-1}, z]$ at fixed order. For a truncated exponential $e^{\sum_{k \geq 1} u_k t^k} = \sum v_j t^j$, use $v_0 = 1$ and $v_j = j^{-1} \sum_{k=1}^j k u_k v_{j-k}$. Finally, differentiation in (25) can be implemented without exponential expressions by iterating

$$e^{-s^2/2} \partial_s (e^{s^2/2} R(s)) = R'(s) + sR(s).$$

The script `reproduce.py` implements precisely these finite recurrences and accepts any nonnegative finite order. Symbolic expression size and runtime grow with the requested order.

As a check on the first correction,

$$P_1(z) = d_1 - \frac{3}{2}z - \frac{\beta}{8}z^2.$$

The first non-Gaussian term in (24) is at order h^2 . Using the Gaussian tilted moments 1, λ , and $1 + \lambda^2$ yields $c_1 = d_1 - 3\lambda/2 - \beta(1 + \lambda^2)/8$, which simplifies to (4).

5 Asymptotic inversion and residual control

An integer sequence has no canonical real inverse. We therefore define the continuous model explicitly and treat the integer threshold separately. Set $\alpha = \log 2$, $\gamma = 3/2$, and $L_0 = \log C$. For fixed $M \geq 0$, let

$$S_M(x) = \sum_{j=0}^M c_j x^{-j/2},$$

$$\mathcal{A}_M(x) = C 2^x x^{-\gamma} e^{\beta\sqrt{x}} S_M(x), \tag{28}$$

$$F_M(x) = \log \mathcal{A}_M(x) = \alpha x + \beta\sqrt{x} - \gamma \log x + L_0 + \log S_M(x). \tag{29}$$

Here and below the model is used only on its eventual positive increasing branch.

Proposition 5.1. *Choose $x_* > 0$ so that*

$$\sum_{j=1}^M |c_j| x_*^{-j/2} \leq \frac{1}{2}, \quad \frac{\gamma}{x_*} + \sum_{j=1}^M j |c_j| x_*^{-j/2-1} \leq \frac{\alpha}{2}. \tag{30}$$

Then $S_M(x) > 0$ and $F'_M(x) \geq \alpha/2$ for $x \geq x_$. For $Y = \log y \geq F_M(x_*)$, there is a unique $x_M(y) \geq x_*$ with $F_M(x_M(y)) = Y$. Any $x_0 \geq x_*$ satisfies the exact residual bound*

$$|x_0 - x_M(y)| \leq \frac{2}{\alpha} |F_M(x_0) - Y|. \tag{31}$$

Proof. Both left sides in (30) decrease to zero, so such a threshold exists and can be found by repeated doubling. The first inequality bounds S_M below by $1/2$. The derivative of (29) is at least

$$\alpha - \gamma/x - \sum_{j=1}^M j |c_j| x^{-j/2-1} \geq \alpha/2,$$

where the positive $\beta/(2\sqrt{x})$ term was discarded. Hence F_M is strictly increasing and tends to infinity. The mean value theorem gives (31) on the interval with endpoints $x_0, x_M(y)$. $\square$

Let $X = Y/\alpha$ and $L = \log X$. The first terms are

$$x_M(y) = X - \frac{\beta}{\alpha}\sqrt{X} + \frac{\gamma}{\alpha}\log X + \frac{\beta^2}{2\alpha^2} - \frac{L_0}{\alpha} + \mathcal{O}\left(\frac{\log X}{\sqrt{X}}\right). \quad (32)$$

For $M \geq 1$, one further term is $X^{-1/2}U_1(L)$, where

$$U_1(L) = -\frac{\gamma\beta}{2\alpha^2}L - \frac{\beta^3}{8\alpha^3} + \frac{\beta(L_0 - 2\gamma)}{2\alpha^2} - \frac{c_1}{\alpha}. \quad (33)$$

With that term included, the error is $\mathcal{O}(X^{-1}\log X)$. For $M = 0$, the same expression uses zero in place of c_1 . For $M \geq 2$, the next polynomial is

$$U_2(L) = \frac{\gamma^2}{\alpha^2}L + \frac{c_1^2/2 - c_2}{\alpha} - \frac{\gamma L_0}{\alpha^2} + \frac{\beta^2\gamma}{2\alpha^3}. \quad (34)$$

Theorem 5.2. *For any fixed $M, J \geq 0$, polynomials $U_1, \dots, U_J$ can be obtained by the recursion below so that, on setting*

$$T_J(X) = X - \frac{\beta}{\alpha}\sqrt{X} + \frac{\gamma}{\alpha}L + \frac{\beta^2}{2\alpha^2} - \frac{L_0}{\alpha} + \sum_{j=1}^J X^{-j/2}U_j(L), \quad (35)$$

we have both

$$F_M(T_J(X)) - \alpha X = \mathcal{O}_{M,J}\left(X^{-(J+1)/2}(1 + \log X)^{J+2}\right), \quad (36)$$

$$x_M(y) - T_J(X) = \mathcal{O}_{M,J}\left(X^{-(J+1)/2}(1 + \log X)^{J+2}\right). \quad (37)$$

The coefficient U_j depends only on $c_1, \dots, c_{\min(j, M)}$. In particular it stabilizes for all $M \geq j$.

Proof. Treat $t = X^{-1/2}$ and L as independent formal symbols. Put

$$v_0 = 1 - \frac{\beta}{\alpha}t + \left(\frac{\gamma}{\alpha}L + \frac{\beta^2}{2\alpha^2} - \frac{L_0}{\alpha}\right)t^2.$$

If $U_1, \dots, U_{j-1}$ have been chosen, set $v_{j-1} = v_0 + \sum_{r < j} U_r(L)t^{r+2}$ and form

$$\begin{aligned} R(v; t, L) &= \alpha t^{-2}(v - 1) + \beta t^{-1}\sqrt{v} - \gamma(L + \log v) + L_0 \\ &\quad + \log\left(1 + \sum_{k=1}^M c_k t^k v^{-k/2}\right). \end{aligned} \quad (38)$$

The t^{-1} and t^0 coefficients vanish for v_0 . Define

$$U_j(L) = -\alpha^{-1}[t^j]R(v_{j-1}; t, L). \quad (39)$$

Adding $U_j t^{j+2}$ to v changes the t^j coefficient only through the first term of (38), by αU_j ; every other change has larger degree. Induction cancels every coefficient through t^J . Each extraction is finite polynomial algebra and involves only c_k with $k \leq j$. In fact $\deg U_j \leq j$: if every coefficient of t^k in $v - 1$ has degree at most $k - 1$ in L , then the same is true of each positive-degree coefficient of an analytic function of v . The square-root term in (38) shifts k to $k - 1$, so its t^j coefficient has degree at most j ; the logarithmic terms have no larger degree. This proves the induction.

Here is a uniform analytic justification of the remainder. Hold real $L \geq 0$ fixed, and let $r_L = \rho/(1 + L)$ for a sufficiently small constant $\rho > 0$. Because the coefficient of t^k in the finite polynomial $v_J - 1$ has degree at most $k - 1$, on the complex disk $|t| \leq r_L$ one has $|v_J - 1| \leq K|t|$. Choose ρ so that this is less than $1/2$ and the argument of the final logarithm in (38) stays within $1/2$ of one. All square roots and logarithms are then analytic on that disk, with their branches fixed at $t = 0$. The apparent poles of R at zero cancel, and on $|t| = r_L$ its magnitude is at most $K'(1 + L)$: the first two terms are of this size, the explicit L term is of this size, and the remaining analytic terms are bounded. All coefficients through degree J vanish. Cauchy's coefficient bound therefore gives, for $0 < t \leq r_L/2$,

$$|R(v_J; t, L)| \leq K'(1 + L) \frac{(t/r_L)^{J+1}}{1 - t/r_L} \leq K'' t^{J+1} (1 + L)^{J+2}.$$

On substituting $L = -2 \log t$, both t and tL tend to zero and this proves (36). Eventually $T_J(X) \geq x_*$. Proposition 5.1 proves (37). Expanding the t^1 coefficient gives (33); the next coefficient is only linear in L , giving the sharper error stated there. $\square$

[write] **Added 2 October 2026, batch 77: an instance of repository results.** The reversion of Theorem 5.2 is an instance of the inversion apparatus of the transseries volume named after Lemma 2.1, and no novelty is claimed for the method. Its dominant block $\alpha x - \gamma \log x = Y$ is the Lambert core `p0:thm:lambert-core` (there $a = \alpha$, $b = -\gamma$), and the remaining terms $\beta\sqrt{x} + L_0 + \log S_M(x)$ are a perturbation of the kind reverted to all orders by `p0:thm:perturbed-inversion`. The article does not use the Lambert normalisation: it expands about $X = Y/\alpha$ in the two symbols $t = X^{-1/2}$ and $L = \log X$, so its $U_j(L)$ are polynomials in $\log X$ rather than functions of a Lambert variable. The residual bound (31) and the analytic remainder argument are the article's own.

6 The integer threshold and rounding

Define $N(y) = \min\{n \geq 0 : a_n \geq y\}$. This is well defined for large y . In fact, Pascal's identity gives

$$a_{n+1} - a_n = \sum_{k=1}^{n+1} \binom{n}{k-1} b(k) > 0 \quad (n \geq 1),$$

since the $k = 2$ term is n and $b(2) = 1$. The main asymptotic also shows $a_n \rightarrow \infty$.

Theorem 6.1. *For each fixed M , write $r = (M + 1)/2$. There is a constant $B_M > 0$ such that for all sufficiently large y , with $X = (\log y)/\log 2$,*

$$\lceil x_M(y) - B_M X^{-r} \rceil \leq N(y) \leq \lceil x_M(y) + B_M X^{-r} \rceil. \quad (40)$$

An asymptotic approximation $T_J(X)$ may replace $x_M(y)$ after enlarging the bracket radius by the bound in (37).

Proof. Theorem 1.1 implies $|\log a_n - F_M(n)| \leq D_M n^{-r}$ for all sufficiently large integers n . Also $x_M(y) \sim X$, and leading-order asymptotics place $N(y)$ in $[X/2, 2X]$ for large y . Within that range $n^{-r} \leq 2^r X^{-r}$, while $F'_M \geq \alpha/2$. Choose $B_M > 2^{r+1} D_M/\alpha$. Every integer $n < x_M(y) - B_M X^{-r}$ in this range satisfies $\log a_n < Y$, and every integer $n \geq x_M(y) + B_M X^{-r}$ satisfies $\log a_n > Y$. Integers outside the range are settled by monotonicity and the leading-order bounds. Taking ceilings proves (40). $\square$

The constants D_M, B_M here are existence constants, not numerically certified thresholds. The real-model residual estimate (31), in contrast, has the explicit factor $2/\alpha$ once (30) is verified. In particular, no unconditional formula $N(y) = \lceil x_M(y) \rceil$ is asserted: an arbitrarily small inverse error can change the ceiling near an integer.

[write] **Added 2 October 2026, batch 77.** This is the staircase mechanism of the same volume, `p0:thm:staircase`: part (1), $N_* = \lceil \nu \rceil$ for an exact monotone interpolation, and part (2), the separation condition, which fails near an integer boundary. Theorem 6.1 brackets $N(y)$ by the inverse of the *model* $\mathcal{A}_M$, which does not interpolate a_n exactly, so the model error enters as the radius $B_M X^{-r}$. The generic arithmetic of parts (1)–(2) is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` (`Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`); those statements concern an arbitrary strictly monotone function and give no formal status to Theorem 6.1.

7 Reproducibility and numerical checks

The companion script has two independent layers: exact integer values of $p(k)$ from Euler’s pentagonal recurrence and exact binomial sums for a_n ; and symbolic truncated-series arithmetic for the coefficients. It verifies the first sixteen OEIS values and symbolically compares c_1, c_2, c_3 with the separately derived formulas. The general-order algorithm was run through $M = 6$; the approximate coefficients are

j	c_j
0	1
1	−4.410009558388217
2	15.80699952226008
3	−55.10689011321474
4	192.6836569593585
5	−685.3858215031009
6	2492.654442088339

The following table gives $a_n/\mathcal{A}_M(n) - 1$, computed from exact integer a_n followed by 90-digit arithmetic. Exact symbolic expressions through c_6 and fuller diagnostics are in `verification.json`.

n	$M = 0$	$M = 3$	$M = 6$
100	-3.238×10^{-1}	2.150×10^{-2}	-9.949×10^{-4}
500	-1.699×10^{-1}	8.020×10^{-4}	-3.421×10^{-6}
1000	-1.252×10^{-1}	1.981×10^{-4}	-2.997×10^{-7}
2000	-9.128×10^{-2}	4.911×10^{-5}	-2.633×10^{-8}
5000	-5.935×10^{-2}	7.802×10^{-6}	-1.060×10^{-9}
10000	-4.257×10^{-2}	1.943×10^{-6}	-9.346×10^{-11}

For $y = a_{10000}$, numerical inversion of $\mathcal{A}_6$ gives $x_6(y) - 10000 \approx -1.3311749344 \times 10^{-10}$. This is a diagnostic; the floating-point root is not an interval-certified calculation, and the displayed numerical signs do not prove a universal error sign. Finite data also do not prove the expansion, which rests on Sections 3–4.

To reproduce the recorded run, use Python 3 with `sympy` and `mpmath` installed:

```
python3 reproduce.py --order 6 --max-n 10000 --output verification.json
pdflatex a292507_asymptotics.tex
pdflatex a292507_asymptotics.tex
```

The TeX source contains the displayed table; the full results are also stored in JSON. The script's `inverse_polynomials(M,J)` function implements (39). The computational work uses no network calls or downloaded sequence tables.

[write] **Added 2 October 2026, batch 77: shipped names.** In the collection the script is `code/reproduce.py`, its recorded output `data/verification.json`, the symbolic inverse check `code/validate_inverse.py` with output `data/inverse_verification.json`, and this source is `article.tex` (delivered as `a292507_asymptotics.tex`). Both scripts write their JSON into the *current* directory (`reproduce.py` by default, `validate_inverse.py` always), so running them with `data/` as working directory would overwrite the shipped records; the report's README gives a replay on a scratch copy. The values in the two tables above equal those of `data/verification.json`.

8 Prior art and limits of the result

The source check on 1 October 2026 found the logarithmic conjecture attributed to Václav Kotešovec on 11 May 2019 and the coefficient formula attributed to Ilya Gutkovskiy on 24 April 2021 in A292507 [1]. A218481 already records the leading binomial-transform asymptotic for partition numbers (2015), and its 2023 comments explicitly give the $e^{r^2/16}$ smoothing factor for inputs proportional to $e^{r\sqrt{n}}n^{-b}$ [2]. The leading constant here is a direct application of that established mechanism after taking the partition difference. A292622 also records a fixed-column asymptotic, which is a different regime from its diagonal A292507 [3].

All-orders expansions and error analysis for the underlying partition function predate this note. Nemes' paper [4], together with the earlier work of Banerjee, Paule, Radu, and Schneider [5], provides relevant context and stronger explicit source-function error estimates than are needed here. The normal moment-generating-function expansion used above is a standard cumulant calculation.

Searches by the sequence identifier, its descriptive title, and combinations of the terms binomial transform, partitions, and asymptotic inverse did not locate a published formula for the specific higher c_j or this inverse. That is a limited negative search result, not evidence of priority or a claim of novelty. This report's contribution is a self-contained verification of the specific expansion, its finite coefficient procedure, and the stated inversion bounds, conditional only on the cited standard partition estimate. It is a Poincaré expansion to arbitrary fixed algebraic order. It does not prove convergence of the resulting a_n correction series or describe a complete exponential transseries; transformed secondary partition sectors would require separate analysis.

References

- [1] OEIS Foundation, *A292507*, contributed by A. P. Heinz, 2017; formula comments by V. Kotešovec (2019) and I. Gutkovskiy (2021). <https://oeis.org/A292507>. Accessed 1 October 2026.
- [2] OEIS Foundation, *A218481*, contributed by P. D. Hanna, 2012; asymptotic formula and comments by V. Kotešovec (2015, 2023). <https://oeis.org/A218481>. Accessed 1 October 2026.
- [3] OEIS Foundation, *A292622*, fixed-column asymptotic comment by V. Kotešovec, 24 October 2018. <https://oeis.org/A292622>. Accessed 1 October 2026.
- [4] G. Nemes, *Simple error bounds for an asymptotic expansion of the partition function*, Ramanujan Journal **65** (2024), 1757–1771. doi:10.1007/s11139-024-00958-8. <https://arxiv.org/abs/2402.07115>; version 2, Lemma 2.2 and its proof.
- [5] K. Banerjee, P. Paule, C.-S. Radu, and C. Schneider, *Error bounds for the asymptotic expansion of the partition function*, Rocky Mountain Journal of Mathematics **54**(6) (2024), 1551–1592. doi:10.1216/rmj.2024.54.1551. <https://arxiv.org/abs/2209.07887>.